AI.305

Cognitive Robotics

COURSE OVERVIEW

Robots with reasoning, memory, and awareness: knowledge, planning, and metacognition in embodied agents.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: THINKING ROBOTS

KEY TOPICS

- What is cognitive robotics
- Cognition vs. reactivity
- Architectures: Soar, ACT-R
- Course roadmap

TEACHING FOCUS

Give robots knowledge and reasoning.

KEY TAKEAWAY Cognitive robots understand what they do.

2

SESSION

60 MINUTES

KNOWLEDGE & REASONING

KEY TOPICS

- Knowledge representation
- Automated reasoning
- Belief revision
- Commonsense reasoning

TEACHING FOCUS

Let robots reason about their world.

KEY TAKEAWAY Reasoning turns data into understanding.

3

SESSION

60 MINUTES

MEMORY & LEARNING

KEY TOPICS

- Episodic and semantic memory
- Lifelong learning
- Memory consolidation
- Robot autobiographical memory

TEACHING FOCUS

Remember and learn over time.

KEY TAKEAWAY Cognition needs memory across experiences.

4

SESSION

60 MINUTES

METACOGNITION & AWARENESS

KEY TOPICS

- Self-monitoring
- Error detection
- Confidence and uncertainty
- Robot self-awareness intro

TEACHING FOCUS

Robots that know what they know.

KEY TAKEAWAY Metacognition enables self-improvement.

COGNITIVE ROBOTICS

5

SESSION

60 MINUTES

HUMAN-LIKE INTERACTION

KEY TOPICS

- Theory of mind for robots
- Collaborative reasoning
- Explainable robot behavior
- Social cognition

TEACHING FOCUS

Interact as a cognitive partner.

KEY TAKEAWAY Partners understand each other's minds.

COGNITIVE ROBOTICS

6

SESSION

60 MINUTES

COGNITIVE ROBOTICS & REVIEW

KEY TOPICS

- Cognitive robot demo
- Course recap and assessment
- Toward bio-inspired robotics (AI.306)
- Reasoning eval

TEACHING FOCUS

Integrate reasoning into a robot.

KEY TAKEAWAY Cognition closes the loop from perception to purpose.